Abdelrahman Khalifa

Software Engineer | Python, CI/CD, Automation, Linux
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SUMMARY

Software engineer who likes building tools and automation for other engineers. Set up multi-architecture CI that tests against real services, built REST backends and firmware-update automation, and writes Python, Rust, C and SQL on Linux. Google Summer of Code 2026 contributor with The Linux Foundation (OpenPrinting) with 11 PRs merged upstream, so used to code review, written design discussion and working with stakeholders in other organizations. Codeforces Expert.

EXPERIENCE

Google Summer of Code Contributor | **The Linux Foundation (OpenPrinting)** | May 2026 – Oct 2026

Open-source project: Printers page for COSMIC Settings, in Rust; passed the final evaluation | *github.com/Abd002/GSOC-2026*

- Built CI in GitHub Actions on x86_64 and aarch64 runners: builds libcups3 from source, starts real CUPS and Avahi services and runs linting and tests against them; added a CI matrix across architectures and CUPS 2/3 to cups-rs.
- Designed the project as 5 Rust crates with a stable API between the UI and a backend service (varlink IPC), so either side can change without breaking the other (cosmic-settings #2182, in review).
- Got **11 PRs merged upstream** through maintainer code review (list), including crash fixes in libcups (C); agreed API changes in writing with mentors from OpenPrinting and System76.

Embedded Software Engineer | **EKSON** | Aug 2025 – Present

Software for 3-DoF and 6-DoF pneumatic motion simulators: backend, desktop app, firmware and CI

- Built a Node.js/Express + MongoDB REST API for remote operation, and automated firmware delivery: new builds uploaded in a web app are downloaded by the desktop app and flashed to devices with esptool; GitHub Actions CI.
- Developed the Python (PyQt5) control app; used asyncio to wait for device and server messages concurrently, cutting CPU use and latency by about 40% (measured).
- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS.

Embedded Linux Systems Intern | **Orange Digital Center (ODC)** | Dec 2024 – Jan 2025

- Built custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; scored 100% in the final evaluation.

OPEN SOURCE PROJECTS

Contributor (all merged) — Zephyr RTOS, Boa JavaScript engine (Rust), Rust Clippy | Mar 2025 – Dec 2025

- Rust Clippy: fixed `ref_as_ptr` false positives in a lint used by Rust developers everywhere (PR #16214); Boa: faster `JsStr::code_points()` (PR #4553); Zephyr RTOS: `IP_MULTICAST_LOOP` socket option (#86977).

PROJECTS

- **Developer environment automation** (Bash): idempotent install script that sets up a full Linux dev environment (packages, configs, editor plugins and language servers); backs up existing files and can be re-run safely, with flags to skip slow steps. GitHub
- **Function Solver and Plotter** (Python, PySide2, SymPy, Matplotlib): input validation, error handling and end-to-end tests with pytest-qt. GitHub
- **Database Systems, CMU 15-445** (C++17, SQL): thread-safe Count-Min Sketch for the BusTub database; SQL analytics in DuckDB (joins, aggregation, upserts). GitHub
- **Inventory Management System** (C#, SQL Server): audit log implemented with SQL Server triggers that record the logged-in user through the session context. GitHub

TECHNICAL SKILLS

Languages: Python (asyncio), SQL, Bash, Rust, C, C++, JavaScript (Node.js)
CI/CD & testing: GitHub Actions (matrix and multi-architecture builds, service setup in CI), automated testing (pytest, Cargo tests, integration tests against real services), linting and formatting checks
APIs & data: REST API design, Express, MongoDB, SQL Server, DuckDB; IPC (varlink)
Engineering practice: Git/GitHub, code review, Linux (Ubuntu, Arch), debugging, technical writing

ACHIEVEMENTS

- Codeforces Expert; 4-time ECPC finalist, 2021–2024; mentor in the local ICPC community (20+ colleagues).

EDUCATION

Bachelor of Science (B.Sc.), Computer and Systems Engineering | **Minia University** | 2020 – 2025

- Grade: Very Good with Honors.